Md Ahsanul Haque, Ph.D.
MAT-120

Homework-3

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Instructions:

- Write your full name (first **and** last name) **inside** the provided box.
- This assignment contains 5 problems. It is **YOUR responsibility** to check that your copy has all 5 problems.
- Make sure to read the instructions for each problem carefully.
- Use correct mathematical notation.
- Write cohesively and follow a logical sequence of steps.
- Check that your answer can be read and understood by the grader.

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1. A water tank in the shape of an inverted right circular cone has a top radius of 4 feet and a height of 12 feet. Water is being pumped into the tank at a constant rate of 10 ft^3/min.
   (a) At what rate is the water level rising when the water in the tank is 6 feet deep?
   (b) At what rate is the wetted surface area of the cone changing at that same instant? (Recall that for a cone with radius r and height h, the lateral surface area is $A = \pi r\sqrt{r^2 + h^2}$).
   (c) At what rate is the volume of water changing with respect to the water height h when $h = 6$ feet?

2. Evaluate each limit using L'Hôpital's Rule (show all indeterminate forms before applying the rule):
   (a) $\lim_{x\to 0} \dfrac{e^{4x} - 1 - 4x}{x^2}$
   (b) $\lim_{x\to 0} \dfrac{1 - \cos(3x)}{x\tan(x)}$
   (c) $\lim_{x\to \infty} \dfrac{\ln(x^3 + 1)}{x}$
   (d) $\lim_{x\to 0^+} (x^2 \ln x)$

3. The concentration of a drug in a patient's bloodstream t hours after injection is given by $C(t) = 150e^{-0.12t}$, where $C(t)$ is measured in mg/L.
   (a) Find the rate at which the drug concentration is decreasing 2 hours after the injection.
   (b) How many hours will it take for the drug concentration to drop to 30 mg/L?
   (c) Find the average rate of change of the drug concentration over the first 4 hours.
   (d) Using L'Hôpital's Rule, evaluate $\lim_{t\to\infty} \dfrac{C(t)+t}{3t+2}$ and interpret its real-world meaning.

4. Consider the polynomial function $f(x) = x^4 - 4x^3 + 10$.
   (a) Find all critical points of $f(x)$.
   (b) Determine the open intervals on which $f(x)$ is increasing and decreasing. Show your work clearly using a sign chart for $f'(x)$.
   (c) Apply the First Derivative Test to classify each critical point as a local maximum, local minimum, or neither.

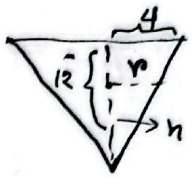
5. Consider the function $g(x) = 3x^5 - 5x^4 + 2$.
   (a) Find the open intervals on which the graph of $g(x)$ is concave up and concave down. Show your work clearly using a sign chart for $g''(x)$.
   (b) Find all inflection points of $g(x)$, providing both x and y coordinates.
   (c) Use the Second Derivative Test, where applicable, to re-verify whether the critical point at $x = 1$ corresponds to a local maximum or local minimum.

Solution to the question no.-1

a) Given, $\frac{dV}{dt} = 10\ ft^3/min$

radius $r = 4\ ft$

height $h = 12\ ft$



$\frac{r}{h} = \frac{4}{12}$ [Similar triangle]

$\therefore r = \frac{h}{3}$

$V = \frac{1}{3}\pi r^2 h$

$\Rightarrow V = \frac{1}{3}\pi\left(\frac{h}{3}\right)^3 h$

$\Rightarrow V = \frac{\pi}{27} h^3$

$\Rightarrow \frac{dV}{dt} = \frac{\pi}{27} \times 3h^2 \frac{dh}{dt}$

$\Rightarrow \frac{dh}{dt} = \frac{\pi}{9} \times \frac{6^2}{h^2} \times \frac{1}{10}$ [$\because$ when $h=6\ ft$; $\frac{dV}{dt} = 10\ ft^3/min$]

$= \frac{10 \times 9}{\pi \times 6^2} = \frac{5}{2\pi}$

$\therefore \frac{dh}{dt} \approx 0.796\ ft/min$

$\therefore$ When the height is 6 ft the water level is rising at $\frac{5}{2\pi}$ or 0.796 ft/min.

b) Given, $A = \pi r(\sqrt{r^2+h^2}$

$r = \frac{h}{3}$

$\therefore A = \pi \frac{h}{3}\sqrt{\frac{h^2}{9}+h^2} = \frac{\pi h}{3}\sqrt{\frac{10h^2}{9}} = \frac{\pi h^2\sqrt{10}}{9}$

Now, $\frac{dA}{dt} = \frac{d}{dt}\frac{\pi h^2\sqrt{10}}{9}$

$= \frac{\pi\sqrt{10}}{9} \cdot 2h\frac{dh}{dt}$

$= \frac{2\pi\sqrt{10}}{9} \times 6 \times \frac{5}{2\pi}$

$= \frac{10\sqrt{10}}{3}\ ft^2/min$

$\therefore$ At 6 ft height the wetted surface area of the cone changing at $\frac{10\sqrt{10}}{3}$ ~~fee~~ square feet per minutes.

c) $V = \frac{\pi h^3}{27}$

$\frac{dV}{dh} = \frac{\pi}{27} 3h^2$

$\Rightarrow \frac{dV}{dh} = \frac{\pi}{9} \times 6^2$

$\therefore \frac{dV}{dh} = 4\pi\ ft^2$

$\therefore$ At 6f height the volume of water is ~~changing~~ rising at $4\pi\ ft^3/ft$ or $4\pi\ ft^2$.

Solution to question no. - 2

a) $\lim_{x \to 0} \frac{e^{4x} - 1 - 4x}{x^2}$; $\frac{0}{0}$

$= \lim_{x \to 0} \frac{4e^{4x} - 4}{2x}$

$= \lim_{x \to 0} \frac{2e^{4x} - 2}{x}$; $\frac{0}{0}$

$= \lim_{x \to 0} \frac{8e^{4x}}{1}$

$= 8$

b) $\lim_{x \to 0} \frac{1 - \cos(3x)}{x \tan x}$; $\frac{0}{0}$

$= \lim_{x \to 0} \frac{3 \sin 3x}{\tan x + x \sec^2 x}$; $\frac{0}{0}$

$= \lim_{x \to 0} \frac{9 \cos 3x}{\sec^2 x + \sec^2 x + 2x \sec^2 x \tan x}$

$= \frac{9}{1 + 1 + 0}$

$= \frac{9}{2}$

$\frac{d}{dx} \sec^2 x$

$= \sec x \cdot \sec x \tan x + \sec x \cdot \sec x \tan x$

$= 2 \sec^2 x \tan x$

c) $\lim_{x\to\infty} \frac{\ln(x^3+1)}{x}$; $\frac{\infty}{\infty}$

$= \lim_{x\to\infty} \frac{\frac{1}{x^3+1} \cdot 3x^2}{1}$

$= \lim_{x\to\infty} \frac{3x^2}{x^3+1}$

$= \lim_{x\to\infty} \frac{\frac{3}{x}}{1+\frac{1}{x^3}}$

$= \frac{0}{1}$

$= 0$

d) $\lim_{x\to 0^+} x^2 \ln x$; $0.(-\infty)$

$= \lim_{x\to 0^+} \frac{\ln x}{\frac{1}{x^2}}$; $\frac{-\infty}{\infty}$

$= \lim_{x\to 0^+} \frac{\frac{1}{x}}{\frac{-2}{x^3}}$

$= \lim_{x\to 0^+} \frac{-x^2}{2}$

$= 0$

Solution to the question no. – 3

a) Given, $c(t) = 150e^{-0.12t}$

$$c'(t) = \frac{d}{dt} 150 e^{-0.12t}$$

$$= 150 \cdot e^{-0.12t} \cdot (-0.12)$$

$$= -18 e^{-0.12t}$$

At, $t = 2$

$$c'(2) = -18 e^{-0.12 \times 2} = -18e^{-0.24}$$

$$= -14.16 \text{ mg/L/Hour}$$

∴ The concentration is decreasing at 14.16 mg per Litre per hour.

b) $c(t) = 30$

$$\Rightarrow 150 e^{-0.12t} = 30$$

$$\Rightarrow e^{-0.12t} = \frac{30}{150}$$

$$\Rightarrow -0.12t = \ln \frac{1}{5}$$

$$\Rightarrow t = \frac{-\ln(\frac{1}{5})}{0.12}$$

$$\approx 13.41 \text{ hour}$$

∴ 13.41 hour will need to drop concentration to 30 mg/L.

c) Average rate of change: $\frac{c(4)-c(0)}{4-0}$

$c(0) = 150\, e^{-0.12t}$

$= 150\, e^{0}$

$= 150$

$c(4) = 150\, e^{-0.12\times 4}$

$= 150\, e^{-0.48}$

∴ Average rate of change: $\frac{c(4)-c(0)}{4}$

$= \frac{150\, e^{-0.48} - 150}{4}$

$= \frac{75}{2}\left(e^{-0.48} - 1\right)$

$= -14.31$ mg/L/hour

∴ The average concentration decreases at 14.31 mg/L/hour.

d) $\lim_{t \to \infty} \frac{c(t) + t}{3t + 2}$

$= \lim_{t \to \infty} \frac{150 e^{-0.12t} + t}{3t + 2}$; $\frac{\infty}{\infty}$

$= \lim_{t \to \infty} \frac{-18 e^{-0.12t} + 1}{3}$ $\left[\because \frac{d}{dt} c(t) = -18 e^{-0.12t}$ we get, from a $\right]$

$= \lim_{t \to \infty} \frac{-18 \times 0 + 1}{3}$

$= \frac{1}{3}$

As time ∞ becomes very large $c(t)$ approaches zero and the ratio approaches $\frac{1}{3}$.

Solution to question no.- 4

a) $f(x) = x^4 - 4x^3 + 10$

$f'(x) = 4x^3 - 12x^2$

Let, $f'(x) = 0$

$\Rightarrow 4x^3 - 12x^2 = 0$

$\Rightarrow 4x^2(x-3) = 0$

$\therefore x = 0$ or, $x - 3 = 0$

$\therefore x = 3$

Substitute x,

$f(0) = 0 - 0 + 10 = 10$

$f(3) = 3^4 - 4 \times 3^3 + 10 = 81 - 108 + 10 = -17$

$\therefore$ Critical points: $(0, 10)$, $(3, -17)$

b)

we have, $f'(x) = 4x^2(x-3)$

critical points at: 0, 3

interval sign check:

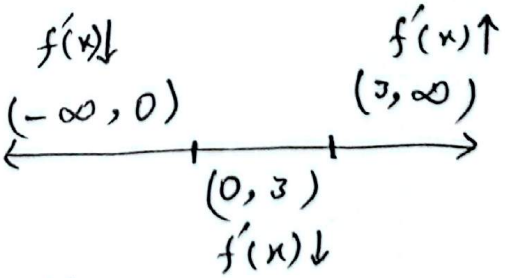
$(-\infty, 0) \Rightarrow f'(-1) = 4(-1)^2(-1-3) = -16$

$(0, 3) \Rightarrow f'(1) = 4 \times 1^2(1-3) = -8$

$(3, \infty) \Rightarrow f'(4) = 4 \times 4^2(4-3) = 64$

$\therefore$ Increasing at $(3, \infty)$

Decreasing at $(-\infty, 0) \cup (0, 3)$



Sign chart

	f'(x)	
$(-\infty, 0)$	−	Decreasing
$(0, 3)$	−	Decreasing
$(3, \infty)$	+	Increasing

c) We get,

$f'(x) = 4x^3 - 12x^2$

critical points at : 0, 3)

interval : $(-\infty, 0)$, $(0, 3)$, $(3, \infty)$

From question b,

	$f'(x)$	
$(-\infty, 0)$	−	$f'(x)<0$
$(0, 3)$	−	$f'(x)<0$
$(3, \infty)$	+	$f'(x)>0$

At $x=0$ there is no sign change before and after.

At $x=3$,

for $x<3$ $f'(x) \Rightarrow -$ or, $f'(x)<0$

for $x>3$ $f'(x) \Rightarrow +$ or, $f'(x)>0$

∴ function changes from increasing to decreasing.

∴ $x=3$ local minimum.

∴ local minimum point, $(3, -17)$

Solution to the question no. 5

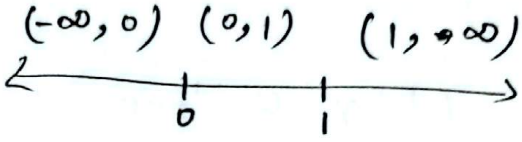
a) Given that,

$g(x) = 3x^5 - 5x^4 + 2$

$g'(x) = 15x^4 - 20x^3$

$g''(x) = 60x^3 - 60x^2$

Let, $g''(x) = 0$



$\Rightarrow 60x^3 - 60x^2 = 0$

$\Rightarrow 60x^2(x-1) = 0$

$\therefore x = 0$ or, $x = 1$

interval check:

$(-\infty, 0) : g''(-1) = 60(-1)^3 - 60(-1)^2 = -120$

$(0, 1) : g''\left(\frac{1}{2}\right) = 60 \times \left(\frac{1}{2}\right)^3 - 60 \times \left(\frac{1}{2}\right)^2$

$= 60 \times \frac{1}{8} - 60 \times \frac{1}{4}$

$= \frac{15}{2} - 15$

$= -\frac{15}{2}$

$(1, \infty) : g''(2) = 60\left(2^3 - 2^2\right)$

$= 60(8-4)$

$= 240$

interval	Sign of $g''(x)$	concavity
$(-\infty, 0)$	−	concave down
$(0, 1)$	−	concave down
$(1, \infty)$	+	concave up

$\therefore$ Concave up on $(1, \infty)$

$\therefore$ concave down on $(-\infty, 0) \cup (0, 1)$

b) we get,

$$g''(x) = 60x^3 - 60x^2$$
$$= 60x^2(x-1)$$

From question a we get, for the interval, $(-\infty, 0), (0,1), (1,\infty)$

interval	Concavity
$(-\infty, 0)$	concave down
$(0,1)$	concave down
$(1,\infty)$	concave up

• At $x=0$,

for $x<0$; concave down

for $x>0$; concave down

So, concavity doesn't change. $x=0$ not a inflection point.

At $x=1$,

for $x<1$; concave down

for $x>1$; concave up

∴ Concavity changes

∴ $x=1$ is a inflection point.

$$f(1) = 3\times 1^5 - 5\times 1^4 + 2$$
$$= 0$$

∴ Inflection point is $(1,0)$.

c) $g'(x) = 15x^4 - 20x^3$

$= 5x^3(3x-4)$

let, $g'(x) = 0$

$5x^3(3x-4) = 0$

$\Rightarrow x = 0$ or, $3x - 4 = 0$

$\therefore x = \frac{4}{3}$

$\therefore$ critical points $0, \frac{4}{3}$

and, $g'(1) = 5(3-4) = -5 \neq 0$

$\therefore x = 1$ is not a critical point. It's inflection point.

Now,

$g''(x) = 60x^2(x-1)$

~~$g''(0) = 60 \times 0(0-1) = 0$~~

$g''\left(\frac{4}{3}\right) = 60 \times \left(\frac{4}{3}\right)^2\left(\frac{4}{3}-1\right)$

$= \frac{960}{27} > 0 \rightarrow$ local minimum

for $g''(0) = 60 \times 0(0-1) = 0$ (minimum/maximum cannot be determined using second derivative test)

Using 1st derivative,

at $x=0$, for $x < 0$, $g'(-1) = 5(-1)^3\{3(-1)-4\} = 35$; $g'(x) > 0$ increasing

for $0 < x < \frac{4}{3}$; $g'(1) = 5(3-4)$

$= -5$; $g'(x) < 0$ decreasing

$\therefore x = 0$ local maximum.

$$g(0) = 0 - 0 + 2$$
$$= 2$$

$\therefore$ local maximum point $(0, 2)$

$$g\left(\frac{4}{3}\right) = 3\times\left(\frac{4}{3}\right)^5 - 5\left(\frac{4}{3}\right)^4 + 2$$
$$= \frac{-94}{81}$$

$\therefore$ local minimum point $\left(\frac{4}{3}, \frac{-94}{81}\right)$